

Motor Selection

* Effective Inertia:

- J_m : motor inertia
- J_L : Limb / Load inertia
- k : spring stiffness
- ω : frequency of iteration

① Direct Drive: *No Compliance; muscle CE < fast* *precise*

$J_m \ddot{\theta} = \tau = J_L \ddot{\theta}$

→ $(J_m + J_L) \ddot{\theta} = \tau \rightarrow J_{eff} = J_m + J_L$

* hit a wall / move slowly → feel combined mass of motor & Limb

* high torque DD is dangerous.

* risky when interacting with humans ↔ if paired with external compliance then it is ok

How Add Compliance →

- ① Hardware: SEA $\left\{ \begin{array}{l} \text{Spring} \\ \text{Elastomer} \end{array} \right.$ → if the robot hit , then the spring first compressed → absorb energy before the motor can cause Damage.
- Compliant end-effector $\leftarrow$ Soft-skin (silicon hand) Flexure → mechanism that bend rather rigid hinges.
- Remote Center Compliance

② Software & Control

- Impedance Control
- Admittance Control

Feature	Impedance Control	Admittance Control
Philosophy	"Motion in --> Force out"	"Force in --> Motion out"
Input	Position/Velocity (from encoders)	Interaction Force/Torque (from F/T sensor)
Output	Torque Command to the motor	Position/Velocity Command to the motor
Analogy	A physical spring you are pushing.	A shopping cart that "admits" your push.
Hardware	Best for "Soft" motors (Direct Drive).	Best for "Stiff" motors (High gear ratios).

The fundamental difference lies in what the controller "sees" as the input and what it "tells" the motor to do.

2. Impedance Control (The "Virtual Spring")

In Impedance control, you treat the robot like a spring-damper system.

- **The Goal:** You want to reach position X_d .
- **The Process:** If an external force pushes the robot away to position X_m , the controller calculates:
$$\tau = K(X_d - X_m) + B(\dot{X}_d - \dot{X}_m)$$
- **The Result:** The motor produces a **Torque** (τ) to push back. You aren't commanding a position; you are commanding a "push" that is proportional to how far away you are from your goal.

3. Admittance Control (Your Beam Project)

What you described for your beam balance is classic **Admittance Control**. You are measuring the "excessive torque" and using it to decide where the beam should go.

- **Your Logic:** You have a desired state (Zero Momentum/Torque Balance). You measure the actual torque (T_{measured}) and find the error ($T_{\text{excess}} = 6 \text{ Nm}$).
- **The Virtual World:** You feed that 6 Nm into a **virtual mass-spring-damper model** in your code.
- **The Command:** The code calculates, "If a 6 Nm force hit a virtual object with this much inertia and damping, it would move at this velocity."
- **The Result:** You send a **Position or Velocity command** to your stiff motor. The motor moves to "admit" the force, which makes the beam feel compliant to the person touching it.

1. The "Square Law" of Reflected Inertia

This is the most important technical reason. When you have a gear ratio (N), the inertia of the motor's rotor is "reflected" to the output shaft by N^2 .

- **For your 100:1 drive:** $100^2 = 10,000$.
- **The Result:** The human at the output shaft doesn't just feel the weight of the robot arm; they feel the motor's internal rotor inertia **10,000 times heavier** than it actually is.
- **Why it kills Impedance:** Impedance control relies on the human being able to move the robot easily so the motor can "feel" that movement (via the encoder) and respond. If the human has to fight 10,000x inertia just to budge the arm, the system is no longer "compliant" or "transparent."

3. Loss of "Transparency"

Impedance control is an **Open-Loop Force** strategy. You are commanding a torque and *assuming* the mechanics will allow the arm to move if someone pushes it.

- With a **Direct Drive**, if I push the arm, the motor spins immediately. The encoder sees the change, and the software says "Oh, he pushed it 1 degree, I should apply X amount of virtual spring force."
- With a **100:1 Harmonic Drive**, if I push the arm with 5 Newtons, the arm **does not move at all** because of the friction and inertia mentioned above. Since the arm doesn't move, the encoder sees **zero change**, and the software thinks no one is touching it.

At a 100:1 ratio, the **reflected inertia** of the motor increases by the square of the gear ratio—meaning the user would feel a virtual mass 10,000 times larger than the rotor itself. Additionally, the **breakaway** mass of the Harmonic Drive is so high that the system is no longer **backdrivable**.

In Impedance control, the motor 'feels' the user through motion (encoders), but here, the gears would mask that motion. I chose **Admittance control** because it uses an external force sensor to bypass the gear friction entirely, allowing the controller to 'see' the user's intent even when the gears are physically locked.

1. The "Bypass": How it physically skips the gears

In a standard robot without a force sensor, the only way the controller knows you are pushing it is if the **motor's encoder** sees the shaft move. With a **100:1 Harmonic Drive**, the gears are so stiff that a human push might not move the motor at all. The encoder sees 0 movement, so the controller does 0 .

The Bypass Solution: You place a **Force/Torque (F/T) Sensor** at the interface where the human touches the robot (the end-effector or the "cuff" of the exoskeleton).

- **Input:** The sensor picks up your 5 N push instantly.
- **The Bypass:** This signal goes directly to the computer **before** the force has to fight the gears. The computer "sees" your intent even though the arm hasn't moved a single millimeter yet.

2. Why it is essential for High Gear Reduction (Reflected Inertia)

You need to mention Reflected Inertia—this is what interviewers are looking for.

When you have a gear ratio of $N=100$, the inertia of the motor (J_m) is "reflected" to the user as $J_m \times N^2$.

- **The Math:** $100^2 = 10,000$. The user feels the motor is **10,000 times heavier** than it actually is.
- **The Stiction:** High-ratio gears have high "stiction" (static friction). You might have to push with 20 N of force just to get the gears to "break away" and start moving.

Admittance Control is essential because it allows the robot to use its own power to overcome its own friction. The robot "feels" your 1 N push and says: "I will start the motor for you so you don't have to fight my 10,000x inertia."

3. The "Translator": From Force to Motion

How does 10 N of force become a "Move 2 degrees" command? You use a **Virtual Model**.

Inside your software, you create a "Ghost Robot" that has no friction. You define its virtual properties:

- **M_v :** Virtual Mass (How heavy you want the robot to feel).
- **B_v :** Virtual Damping (How "thick" or "viscous" the air feels).
- **K_v :** Virtual Stiffness (usually set to 0 for "transparent" following).

The Calculation Loop:

1. **Read Force:** $F_{\text{ext}} = 10 \text{ N}$.
2. **Calculate Acceleration:** $\ddot{x} = \frac{F_{\text{ext}}}{M_v}$.
3. **Integrate for Velocity:** $\dot{x}_{\text{new}} = \dot{x}_{\text{old}} + (\ddot{x}) \cdot \Delta t$.
4. **Integrate for Position:** $x_{\text{new}} = x_{\text{old}} + (\dot{x}_{\text{new}}) \cdot \Delta t$.

4. The Execution: The Inner Loop

Once you have the x_{new} (the target position), you send it to the motor's **Internal Position Controller** (usually a high-gain PID loop).

Because the motor is very strong and the PID is "stiff," the motor will move to that position accurately, regardless of the gear friction. To the human, it feels like the robot is "weightless" and following their hand perfectly.

☑ Summary for an Interview:

"The 'bypass' happens because we sense the user's force at the point of contact **before** it enters the transmission. In a 100:1 system, the reflected inertia and friction would otherwise lock the joint. By feeding that force into a **virtual mass-damper model**, we calculate the motion a 'perfect' frictionless robot would have, and then we command our actual motor to reach that position using a high-gain inner loop."

Why Not Impedance for high Gear Reduction

ADMITTANCE Control

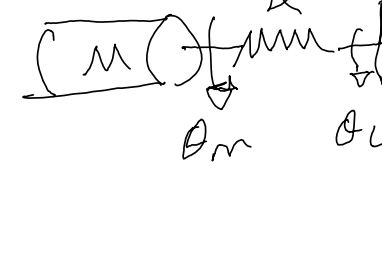
Motor Selection

* Effective Inertia:

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② SEA:

→ Spring is added between Motor & Load:


$$J_m \ddot{\theta}_m - k(\theta_m - \theta_L) \rightarrow J_L \ddot{\theta}_L = k(\theta_m - \theta_L) + \tau_L$$

$$(J_m s^2 + k) \theta_m - k \theta_L \rightarrow \theta_m = \frac{k}{k + J_m s^2} \theta_L$$

$$J_L \ddot{\theta}_L + k \left(\frac{k}{k + J_m s^2} \right) \theta_L + k \theta_L = \tau_L$$

$$\rightarrow \left(J_L s^2 + k \left[\frac{k - k + J_m s^2}{k + J_m s^2} \right] \right) \theta_L = \tau_L$$

$$\rightarrow s^2 \theta_L \left(J_L + \frac{k J_m}{k + J_m s^2} \right) = \tau_L$$

$$J_{eff} = J_L + \frac{k J_m}{k - J_m s^2}$$

$\omega = 0 \rightarrow J_{eff} = J_L + J_m \Rightarrow$ spring is stiff enough to move the motor

$\omega = \infty \rightarrow J_{eff} = J_L \Rightarrow$ high speed impact $\rightarrow$ spring decouple $\rightarrow$ motor $\rightarrow$ environment just feel the light limb.

1. Direct Drive (DD) & Quasi-Direct Drive (QDD)

In this setup, a high-torque motor is connected directly (or through a very low-ratio gearbox) to the joint.

- **Pros:**
 - **High Control Bandwidth:** Because there is no spring "slop," the motor can react instantly. This is why DD robots feel very "crisp."
 - **Transparency/Backdriveability:** If you push the robot arm, the motor spins easily. The robot "feels" the world perfectly because there is no friction or spring hiding the external force.
 - **Simplicity:** No springs to break or calibrate.
- **Cons:**
 - **High Effective Inertia:** As we calculated, the motor's mass is always "visible" to the world. If the robot hits a wall, the motor's rotor inertia slams into the wall too.
 - **Low Torque Density:** To get enough torque without a gearbox, motors must be very large and heavy.
 - **Fragility:** High-speed impacts can shatter the motor magnets or bearings because there is no mechanical shock absorber.
- **Best For:** Small quadrupeds (MIT Cheetah) where high-speed leg swinging is more important than carrying heavy loads.

2. Series Elastic Actuators (SEA)

The "Classic" approach for safe humanoids. The motor moves a spring, and the spring moves the joint.

- **Pros:**
 - **Mechanical Low-Pass Filter (Safety):** At high frequencies (impacts), the spring absorbs the energy before it reaches the motor. As we derived, $J_{eff} \approx J_{load}$ during a hit. This protects both the robot and the human.
 - **Force Sensing:** By measuring the spring's deflection, you have a built-in "Force Sensor." You don't need expensive strain gauges; you just need to know how much the spring bent.
 - **Energy Storage:** Like a human tendon, the spring can store energy during one phase of a gait and release it in the next.

The "Passive" Protection

The most important benefit is that the spring is **passive**. It doesn't need electricity or a computer to work. If a **1X Neo** robot accidentally hits a person's hand, the spring compresses in **0.1 milliseconds**. A 1kHz control loop takes **1.0 millisecond** just to "think." The spring protects the human (and the robot's expensive gears) before the software even knows a collision happened.

- **Cons:**
 - **Lower Bandwidth:** The spring introduces "lag." You cannot move the joint faster than the resonant frequency of the spring-mass system.

The Mechanical Low-Pass Filter

The spring-mass system has a natural resonant frequency (f_n):

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{J_L}}$$

- **k :** Spring stiffness.
- **J_L :** Load inertia (the weight of the arm).

If you try to command the robot to move at a frequency higher than f_n , the spring just absorbs the motion rather than passing it to the arm. This is why we say the spring acts as a **low-pass filter**.

- **Direct Drive Bandwidth:** Can be **50–100 Hz** or higher.
- **SEA Bandwidth:** Often limited to **10–20 Hz**.

For a humanoid, a 10 Hz bandwidth is fine for walking, but it's too slow for "cat-like" reflexes or catching a falling object.

- **Stability Issues:** Controlling an SEA at 1kHz is harder than a DD because the spring can cause oscillations if your control gains aren't tuned perfectly.
- **Best For:** Humanoids in home environments (1X Neo, Agility Digit). Safety is the #1 requirement here. The primary disadvantage of an SEA is that the mechanical compliance limits the **controllable bandwidth**. Because the motor and load are separated by a spring (non-collocated), high-frequency control gains can trigger **structural resonances**, leading to instability. We sacrifice this high-speed precision to gain **passive shock tolerance** and high-fidelity force sensing, which are critical for safe human-robot interaction.

1. What bandwidth means here (precisely)

When people say:

"SEAs have lower bandwidth"; they are **not** talking about sensor bandwidth or controller sampling rate.

They mean: **The fastest frequency at which joint torque can be accurately controlled and modified.**

In other words: How quickly can the actuator **change force/torque at the joint** in response to a disturbance?

This is a **physical limit**, not a software one.

3. Why does the motor need to "feel the world"?

This is the most important conceptual point. **Control is not about knowing force**; It's about **reacting fast enough to shape interaction**.

You asked: "If I know joint position and force, why does the motor need to feel it?" Because: **Force sensors tell you what happened. Motor transparency determines whether you can prevent what happens next.**

Human analogy (very important)

In humans:

- Muscle spindles (velocity)
- Golgi tendon organs (force)
- Reflexes act in **30–50 ms**
- Passive tissue stiffness acts at **0 ms**

But crucially: Muscles are **directly coupled** to joints. There is **no compliant spring isolating the nervous system** from the joint. So force propagates **immediately** into muscle tension.

SEA case

With an SEA:

- Joint $\rightarrow$ Spring $\rightarrow$ Motor
- Fast disturbances are **absorbed by spring deflection**
- Motor torque changes **after** deflection
- Reflex-like response is delayed

So even if you:

- Measure spring force perfectly
- Sample at 10 kHz
- Have perfect control laws

You are still limited by:

$$\tau_{response} \geq \tau_{\omega n} \tau_{response} \geq \frac{1}{\omega_n} \tau_{response} \geq \omega_n$$

A spring deflects at all frequencies, but **at low frequency the motor follows the load so deflection stays small**, while **at high frequency the motor inertia prevents motion so the spring takes nearly all the relative motion and stores the disturbance energy**, mechanically decoupling the motor from fast contact dynamics.

3. Parallel Elastic Actuators (PEA)

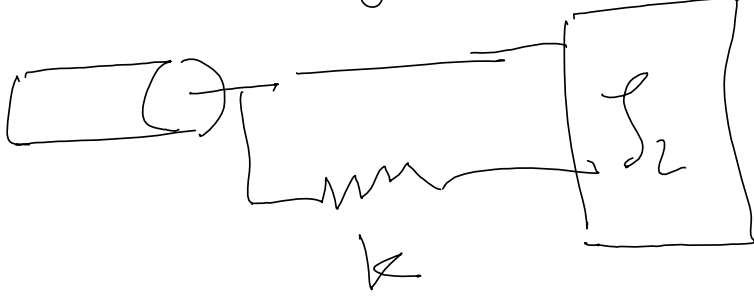
In a PEA, a passive spring is added "next to" the motor. The motor and spring work together to move the same load.

- **Pros:**
 - **Gravity Compensation:** If a robot is standing, the PEA spring can be tuned to "hold up" the robot's weight. The motor then uses **zero power** just to stand still.
 - **High Peak Torque:** If the motor and spring both pull at the same time, you get more torque than the motor could provide alone.
 - **Efficiency:** Excellent for cyclic tasks (like the "bounce" of a bipedal run).
- **Cons:**
 - **No Impact Protection:** Unlike the SEA, the motor is still rigidly connected to the joint. A hit to the arm still goes straight to the motor's gears.
 - **Asymmetric Effort:** A spring that helps you "stand up" will actively "fight" you when you try to "squat down."
- **Best For:** Heavy-duty bipeds or prosthetics where battery life and weight-carrying are the main goals.

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$$\begin{aligned} \tau_{ext} &= J_L \ddot{\theta} + J_m \ddot{\theta} + k \theta \\ &= (J_m + J_L) \ddot{\theta} + k \theta \\ &= (J_m + J_L) s^2 \theta + k \theta \\ &= [(J_m + J_L) s^2 + k] \theta \end{aligned}$$

$$\Rightarrow \frac{\tau_{ext}}{s^2 \theta} = (J_m + J_L) + \frac{k}{s^2}$$

1. The Torque Balance Equation

Imagine a robot joint (like a knee) holding up a leg. There are three main torques acting on that joint:

- τ_g (Gravity):** The weight of the leg pulling it down.
- τ_s (Spring):** The force of the spring pushing it up.
- τ_m (Motor):** The effort the motor has to put in to maintain the position.

The total torque required to stay in a specific position θ is:

$$\tau_{total} = \tau_g(\theta)$$

In a PEA, this total torque is shared between the motor and the spring:

$$\tau_m + \tau_s(\theta) = \tau_g(\theta)$$

To achieve Gravity Compensation, we want the motor effort to be zero ($\tau_m = 0$). This happens when:

$$\tau_s(\theta) = \tau_g(\theta)$$

2. The Intuitive Formula

Let's look at the formulas for gravity and a linear spring to see how they "cancel" each other out.

Gravitational Torque (τ_g):

If a robot arm has mass m and length L , the torque caused by gravity at an angle θ is:

$$\tau_g = m \cdot g \cdot L \cdot \sin(\theta)$$

As the arm moves from vertical to horizontal, the torque increases.

Spring Torque (τ_s):

For a simple torsional spring with stiffness k and a "rest position" (where it exerts no force) at θ_0 :

$$\tau_s = k \cdot (\theta_0 - \theta)$$

The Compensation Goal:

To hold the arm up without using the motor, we set them equal:

$$k \cdot (\theta_0 - \theta) \approx m \cdot g \cdot L \cdot \sin(\theta)$$

3. Why it is so efficient

In a standard robot (Direct Drive), if the robot is just standing still, the motor is constantly drawing current to fight τ_g . This generates heat and drains the battery.

In a PEA:

- Static Holding:** If you pick a spring where k matches the weight of the limb at a specific angle (like a standing pose), the spring provides **100% of the torque**.
- The Motor's Job:** The motor only provides the **difference** $(\tau_g - \tau_s)$. It only works when you move *away* from the equilibrium point or when you need to move very fast.

4. The "Cons" Explained: Why you "Fight" the spring

You mentioned that a spring that helps you "stand" will "fight" you when you "squat." This is the **Asymmetric Effort** problem.

- Standing up:** Gravity wants to pull you down. The spring wants to pull you up. They help each other. τ_m is very small.
- Squatting down:** Gravity wants to pull you down. But now, **the spring also wants to pull you up**. To squat, the motor must push *against* the spring to force the joint to bend.

In a PEA, you are essentially "tuning" the robot to be very efficient at **one specific task** (like standing or walking) at the cost of making other movements harder.

3. Physical Interpretation for your Interview

When an interviewer looks at this equation, they want to see if you understand the **frequency-dependent behavior**.

- At High Frequencies ($\omega \rightarrow \infty$): The term $\frac{k}{s^2}$ goes to zero. $J_{eff} \approx J_m + J_L$.

Meaning: If you shake the robot very fast, the spring doesn't have time to help. You feel the full weight of both the motor and the load.

- At Low Frequencies ($\omega \rightarrow 0$): The $\frac{k}{s^2}$ term becomes very large. The effective inertia becomes a large negative number.

Meaning: This is the "Gravity Compensation" effect. If you move slowly, the spring torque is so strong that it "pushes" the load for you. It's like the robot has "Negative Mass"—it wants to accelerate on its own because of the stored spring energy.

- At Resonance ($\omega = \sqrt{\frac{k}{J_m + J_L}}$): $J_{eff} = 0$.

Meaning: At this specific frequency, the system is perfectly balanced. You could move the robot with almost zero effort because the spring and inertia are perfectly canceling each other out.

Actuator Type	Configuration	Primary Benefit	Jeff Behavior
SEA (Series)	Motor --> Spring --> Load	Safety & Impacts	Decouples J_m at high frequencies.
PEA (Parallel)	Motor Spring -> Load	Energy Efficiency	Subtracts inertia at low frequencies (Gravity Comp).

While PEAs are great for reducing the average torque the motor needs to produce (energy efficiency), the **SEA (Series Elastic)** approach is what allows Neo to have such low effective inertia at high frequencies. This makes the robot physically safe by design, as the motor's mass is effectively 'hidden' from the user during accidental collisions.

Why this matters for your Humanoid/Biped/Quadruped:

For a Humanoid (e.g., 1X Neo):

You want SEAs in the arms and legs.

- **Why?** Because a humanoid interacts with humans. If Neo accidentally hits a child's hand, the **Series Spring** allows the limb to "give" instantly, even before the 1kHz software can react. It mimics the **compliance of the human muscle-tendon unit**.

For a Biped (e.g., Balancing/Walking):

You might use a Hybrid (SEA + PEA).

- **SEA** in the ankle for fine balance control and impact absorption.
- **PEA** in the knee to help support the weight of the torso without draining the battery.

For a Quadruped (e.g., Running/Jumping):

Usually **QDD (Quasi-Direct Drive)**.

- **Why?** Running requires extremely fast leg repositioning. A spring (SEA) might be too slow to "swing" the leg forward in time for the next step. Quadrupeds often rely on "Software Compliance" (simulating a spring using high-speed motor control) rather than a physical spring.

Key "Interview Script" for Christophe:

*"While Direct Drive offers high transparency, it lacks the 'Passive Safety' inherent in biological systems. By using **Series Elastic Actuators**, we replicate the **Hill Muscle Model**, effectively decoupling the motor inertia from the environment during high-frequency impacts. This allows us to achieve 'Human-Safe' interaction without sacrificing the force-density needed for heavy lifting, especially if we augment it with **Parallel Elasticity** for gravity compensation in the stance phase."*

Large scale Robot:

This is a classic "Systems Design" interview question. They aren't just looking for one type of actuator; they want to see if you understand the **trade-offs of scale** and the **physics of environment**.

If I were answering this in an interview, I would frame the answer using three pillars: **The Square-Cube Law (Scaling), Actuator Topology (SEA vs. QDD vs. PEA), and Environmental Robustness**.

1. The Challenge of Scale: Small Robot vs. "Horse" Scale

When you move from a small 3G1 robot to a "Horse" sized humanoid, physics works against you. This is the **Square-Cube Law**.

- **Mass (\$m\$):** Increases by the **cube** of the scale (L^3).
- **Strength/Torque (\$\tau\$):** Increases by only the **square** of the scale (L^2) (cross-sectional area of the motor/gears).
- **Inertia (\$J\$):** Increases by the **fifth power** (L^5)!

The General Conclusion: As you scale up, your robot becomes "lazier" and "weaker" relative to its weight.

- **Small Robot (Agile):** You can use **Quasi-Direct Drive (QDD)**. Because the mass is low, you don't need high gear ratios. This gives you incredible agility and "transparency."
- **Large Robot (Heavy Work):** You are forced to use **High Gear Ratios (e.g., 100:1 or more)** to get the torque needed to lift its own heavy limbs. This introduces high reflected inertia, making the robot "stiff" and prone to breaking its gears upon impact.

2. Choosing the Actuator Strategy

For a hazardous construction zone, I would propose a **Hybrid Actuation Strategy** rather than picking just one.

A. The Lower Body (Legs): Series Elastic Actuators (SEA)

- **Why:** Construction sites are uneven. The robot *will* trip, fall, or drop heavy beams.
- **The Benefit:** In a large-scale robot, the gears are the most fragile part. An SEA provides **passive mechanical protection**. When the robot's foot hits a concrete step hard, the spring absorbs the shock before it shatters the high-ratio harmonic drive.
- **Efficiency:** You can use the SEA springs to store energy during walking (like a human Achilles tendon), which is vital for a heavy robot's battery life.

B. The "Back/Waist": Parallel Elastic Actuators (PEA)

- **Why:** Manual labor involves holding heavy loads for long periods.
- **The Benefit:** If the robot is carrying a 50kg bag of cement, a PEA (a spring in parallel with the motor) can "cancel out" that weight. The motor only provides the torque to move, while the spring provides the torque to "hold." This prevents thermal shutdown of the motors in a hot construction environment.

C. The Upper Body/Arms: Quasi-Direct Drive (QDD)

- **Why:** Agility and Interaction.
- **The Benefit:** If the robot needs to catch a falling tool or react to a human coworker, it needs high **bandwidth**. QDDs (low gear ratio, high-torque motors) allow the arms to be "transparent" and fast.

3. Design Considerations for Hazardous Zones

Since you mentioned a "hazardous construction zone," three extra things must be in your answer:

- **Thermal Management:** Large motors generate massive heat, especially when fighting high friction in geared systems. I would implement **Liquid Cooling** or advanced heat-sinking for the leg actuators, as they do the most work.
- **Ingress Protection (IP Rating):** Construction sites are dusty (concrete dust is abrasive) and wet. Actuators must be **sealed** (IP65 or higher). This is a "pro" for SEAs because you can seal the motor/gearbox inside a housing and have only the spring-load interface exposed.
- **Admittance Control for Manual Labor:** Since the large robot has high-ratio gears, I would use **Admittance Control** for the arms. This allows the robot to feel a human "leading" it (like a coworker helping carry a heavy pipe) without the user having to fight the motor's internal friction.

4. Summary for the Interviewer

"To design an agile, large-scale construction humanoid, my primary consideration is managing the **L^5 scaling of inertia**.

I would move away from the **Quasi-Direct Drive** used in small robots and implement **Series Elastic Actuators (SEA)** in the legs to protect the high-ratio gears from the high-impact shocks typical of construction zones. To handle heavy manual labor, I would add **Parallel Elasticity (PEA)** to the lumbar and knee joints to offload the static weight, improving energy efficiency. Finally, I would use an **Admittance Control** framework to ensure that despite the high gear ratios required for scale, the robot remains safe and 'compliant' when interacting with human workers."

NEO Actuators:

The 1X Neo's actuation system is unique because it doesn't fit neatly into a single category. Instead, it is best described as a **Tendon-Driven Proprioceptive (Quasi-Direct) Drive**.

1X Technologies intentionally rejected the "Series Elastic Actuator" (SEA) path for their humanoid for the same reason it's a "Con" in the previous list: **Bandwidth**. If you want a robot to catch a ball or react at lightning speed, a physical spring (SEA) is often too "mushy."

Here is the technical breakdown of how Neo operates compared to the other architectures:

1. The Motor: Revo1 (High-Torque Direct Drive)

At the heart of every Neo joint is the **Revo1** motor. While most humanoid motors are "off-the-shelf," 1X built their own to have the world's highest torque-to-weight ratio.

- **Relationship to DD:** The motor itself is a **Direct Drive** motor. It is designed to produce massive torque at low speeds so it doesn't need a heavy, high-ratio gearbox (like a 100:1 Harmonic Drive) that would ruin backdriveability.

2. The Transmission: The "Tendon Drive" (Cable-Based)

Instead of using gears or belts, 1X uses a **patented Tendon Drive**.

- **Mechanism:** High-strength ropes (tendons) wrap around the motor and the joint.
- **The "Hybrid" Nature: * Like an SEA:** The ropes have a tiny bit of inherent "give" (compliance). While they aren't bouncy springs, they act as a high-frequency shock absorber, protecting the motor magnets from impact.
 - **Like a QDD:** The "gear ratio" is very low. This makes the system **95% backdriveable**. If you push Neo's arm, you aren't fighting a gearbox; you are spinning the motor rotor directly through the ropes.

3. Safety: Soft Body + Virtual Compliance

Neo achieves the "Safety" of an SEA without the "Lag" of a physical spring through two layers:

- **Passive Safety (External):** Neo's "muscles" are encased in a **3D-lattice polymer (soft body)**. If it hits you, the outer shell deforms. This is a "Parallel Elastic" effect, **but for the skin, not the joint**.
- **Active Safety (Proprioception):** Because the Revo1 motor is so transparently connected to the joint, the robot can "feel" a single gram of pressure on its hand by measuring the electrical current in the motor. This is called **Proprioceptive Sensing**.

Feature			
Classic SEA (e.g. Digit)			
Direct Drive (e.g. MIT Cheetah)			
1X Neo (Tendon Proprioceptive)			
Feature,	Classic SEA (e.g. Digit),	Direct Drive (e.g. MIT Cheetah),	1X Neo
Compliance, Physical Spring (High lag),	Software Only (None),	Hybrid (Tendon give + Software)	
Reaction Speed,	Low (Spring oscillation),	High (Crisp),	High (Crisp)
Force Sensing,	Measuring spring bend,	Measuring motor current,	Measuring motor current (very sensitive)
Impact Safety,	Mechanical,	Electronic (Too slow?),	Mechanical (Rope + Soft Shell)
Complexity,	High (Spring + 2 Encoders)	Low,	Moderate (Rope tensioning)

1) “Proximal to what?” (in tendon-driven actuation)

Proximal means **closer to the robot’s body / torso**, not at the joint itself.

In biology

- **Muscles** are usually located **proximally** (e.g., in the thigh or forearm)
- **Tendons** transmit force **distally** to the joint (knee, ankle, wrist, fingers)

Example:

- Calf muscles (proximal, in the shank)
- Achilles tendon (distal transmission)
- Ankle joint (where torque is applied)

This reduces:

- distal mass
- distal inertia
- swing effort

In Neo’s tendon-driven system

- The **motor** is placed:
 - closer to the torso
 - or closer to the joint axis but not directly inline like a gearbox
- The **rope/tendon** routes force to the joint pulley

So when we say:

“allows proximal actuator placement”

We mean:

- The motor does **not have to sit exactly at the joint center**
- Heavy components can be moved:
 - closer to the trunk
 - away from distal links (hands, forearms, feet)

Why this matters biomechanically

- Lower distal inertia → safer impacts
- Faster swing motions
- Lower energy cost
- More human-like mass distribution

This is **directly inspired by musculoskeletal anatomy**.

2) “Micro-compliance for shock filtering” — how exactly?

Important distinction:

- This is **not** a Series Elastic Actuator
- There is **no intentional spring**
- The compliance is **small, passive, and high-frequency**

Where the micro-compliance comes from

a) Rope material elasticity

High-strength synthetic ropes (e.g., Dyneema-class materials):

- Are very stiff
- But **not infinitely stiff**
- Have small elastic strain under load (sub-millimeter scale)

This creates:

- very small deflection
- only at **high-frequency impacts**

b) Distributed compliance

Unlike a single steel gear mesh:

- The rope wraps around pulleys
- Load is distributed across contact length
- This slightly spreads impulse forces over time

c) No backlash, no gear teeth

Gear impacts are impulsive:

- tooth-to-tooth contact
- sharp torque spikes

Rope drives:

- continuous contact
- smoother force transmission

What kind of shocks does this filter?

- Very fast, high-frequency disturbances:
 - accidental taps
 - small collisions
 - impulsive contact at the hand
- It does **not** act as a low-frequency compliance (like an SEA)

Think of it as:

A mechanical low-pass filter at very high frequencies

Biological analogy

- Tendons are:
 - very stiff at low frequencies
 - slightly compliant at high strain rates
- This protects:
 - muscles
 - joints from impulsive loads

Neo's tendon drive does the **same thing**, but at a robotic scale.

3) Active Proprioceptive Compliance — *how exactly does it work?*

This is the most important concept.

Core idea

Neo does **not rely on hardware springs** to be compliant. It relies on **motor transparency + sensing + control**.

Step-by-step mechanism

1) High backdriveability (mechanical prerequisite)

Because:

- low gear ratio
- no harmonic drive
- rope transmission

External forces at the joint:

- propagate back to the motor
- cause measurable motor torque

This is essential. Without backdriveability, nothing below works.

2) Torque sensing via motor current (proprioception)

In electric motors:

- Torque ∝ current

Because the transmission is transparent:

- Joint torque ≈ motor torque
- Motor current becomes a **virtual force sensor**

This is called **proprioceptive sensing**:

- sensing force internally
- without external force sensors

This is biologically analogous to:

- Golgi tendon organs
- muscle spindle feedback

3) High-bandwidth torque control

Because there is no physical spring:

- torque commands affect the joint almost immediately
- no spring dynamics
- no phase lag from elastic elements

The controller can:

- detect unexpected contact
- adjust torque within milliseconds

4) Virtual impedance / admittance control

Software defines:

- stiffness
- damping
- torque limits

And can change them **instantly**, depending on context:

- free motion → low impedance
- manipulation → moderate impedance
- collision → rapid torque reduction

This is **active compliance**, not passive.

Why this beats SEAs for humanoids

SEAs:

- compliance is fixed by spring
- good for safety
- bad for bandwidth and precision

Neo's approach:

- compliance is **context-dependent**
- safety without lag
- fast reactions (catching, balancing, manipulation)

One clean sentence you can use in the interview

“Neo achieves compliance through mechanical transparency and high-bandwidth proprioceptive torque control rather than physical springs, allowing it to be soft when needed and stiff when required, similar to biological reflex modulation.”

Ultra-short summary (memorize this)

- **Proximal** → closer to the torso, reducing distal inertia
- **Micro-compliance** → tiny rope elasticity filters high-frequency shocks
- **Active compliance** → backdriveable motor + current sensing + software impedance

Proprioception

The Lecturer's Desk – What is Proprioception?

"Proprioception" is often called our 'sixth sense.' While vision tells you where the door is, proprioception tells you where *you* are without you having to look at your limbs. In a biological system, this isn't just a static "map"; it is a high-speed feedback loop consisting of three main variables: Length, Velocity, and Force.

To understand this, imagine your muscles are equipped with two different types of "sensors":

- Muscle Spindles (Length & Velocity):** These are buried inside the muscle fibers. They act like **linear encoders**. They detect how much a muscle is stretched (length) and how fast it is stretching (velocity). If you trip, these sensors detect the sudden stretch and trigger a reflex to stiffen the muscle before your brain even knows what happened.
- Golgi Tendon Organs (Force/Tension):** These are located where the muscle meets the bone (the tendon). They act like **strain gauges**. They measure the pulling force the muscle is exerting. If the force is too high (e.g., trying to lift a car), they can actually "shut off" the muscle to prevent it from tearing.

In your words: You mentioned muscle length, velocity, and force are changing constantly. You are exactly right. Human locomotion isn't a pre-programmed "video"; it's a **reactive dance**. Every step adjusts based on these three variables to maintain balance.

Biological Components of Proprioception

1. Muscle Spindles

- Measure **muscle length** and **rate of stretch**
- Provide:
 - position sense
 - velocity feedback
- Enable reflexes like stretch reflex

2. Golgi Tendon Organs (GTOs)

- Measure **muscle force / tendon tension**
- Protect muscles from excessive load
- Enable force regulation

3. Joint Receptors

- Sense joint angle extremes
- Provide coarse positional awareness

Humans do not need vision to stand or walk

→ proprioception is the dominant feedback loop

2. Translating Proprioception into Robotics

Core question:

How does a robot "feel itself" without skin or muscles?

Robotic Proprioception means:

Internal sensing of:

- joint position
- joint velocity
- joint torque
- actuator state

without relying on external sensors like:

- force plates
- tactile skins
- vision

3. The Key Insight: Proprioception Requires Transparency

Biological analogy

- Tendon tension maps directly to muscle force
- No gears
- No backlash
- No hidden dynamics

Robotic requirement

To achieve proprioception:

External forces must propagate back to the actuator

This is the single most important principle.

What breaks proprioception?

High gear ratios (e.g., Harmonic Drives)

- External forces do NOT reach motor
- Backdriveability is poor
- Motor "doesn't feel" the world

Result:

- You cannot infer joint torque from motor current
- You lose force awareness

5. How Proprioception Is Achieved in 1X Neo

Now we get specific.

5.1 The Motor: Rev01 (Proprioceptive Actuator Core)

Design goal

- Highest torque-to-weight ratio
- Direct torque sensing via current
- Minimal gearing

Why this matters

Motor current obeys:

$$\tau = K_t \cdot I \Rightarrow I = \tau / K_t$$

Where:

- τ = motor torque
- K_t = torque constant
- I = current

Because transmission is transparent:

$$\tau_{\text{joint}} = r \cdot \tau_{\text{motor}} \Rightarrow \tau_{\text{motor}} = \tau_{\text{joint}} / r$$

So:

Joint force is observable internally

This is robotic proprioception.

5.2 The Tendon Transmission (Force Transparency)

Why not gears?

Gears:

- reflect inertia
- filter force
- create backlash

Tendon advantages

- Continuous force path
- No discrete tooth contact
- High mechanical transparency

Biological analogy

- Tendons transmit force with minimal loss
- Neo's ropes do the same

This ensures:

- motor "feels" joint interaction
- external contact → motor current change

5.3 What Neo Can Sense Proprioceptively

Without external sensors, Neo can estimate:

Quantity

How

Joint position

Encoder

Joint velocity

Encoder differentiation

Joint torque

Motor current

Contact force

Torque deviation

Collision

Current spike

Weight of object

Sustained current

Human touch

Small torque disturbance

This is **muscle-like sensing**.

6. Active Proprioceptive Compliance (Key Concept)

Definition

Active compliance is software-defined compliance using:

- proprioceptive sensing
- real-time control

Instead of physical springs.

How it works step-by-step

Step 1: Detect force via current

Unexpected torque → current spike

Step 2: Interpret force

Is this:

- intentional manipulation?
- contact?
- collision?
- gravity load?

Step 3: Modulate impedance

Controller adjusts:

- stiffness
- damping
- torque limits

In real time (milliseconds).

This mimics biological reflex loops

Biology

Neo

Muscle spindle

Encoder + velocity

Golgi tendon organ

Motor current

Reflex modulation

Software impedance

CNS control

Central controller

7. Why 1X Rejected Series Elastic Actuators

SEAs give:

- passive safety
- but fixed compliance
- bandwidth loss
- phase lag

Humanoids require:

- fast balance recovery
- manipulation
- catching
- reflex-level response

Neo achieves:

- SEA-like safety**
- without SEA lag**

By:

- soft outer body
- tendon micro-compliance
- proprioceptive control

8. Proprioception + Soft Body = Whole-Body Safety

1X uses **layered safety**:

Passive (external)

- Polymer lattice skin
- Energy absorption

Active (internal)

- Proprioceptive sensing
- Torque limiting
- Reflex shutdown

This mirrors:

- skin + muscle + reflex arc in humans

9. Why This Matters for Household Humanoids

Household robots must:

- touch humans
- interact unpredictably
- be quiet
- be safe
- be fast

Proprioception enables:

- intuitive interaction
- no violent motion
- smooth response
- trust

Without it:

- robot feels "dead"
- interaction feels dangerous

10. Interview-Ready One-Liners (Use These)

Short:

"Proprioception in Neo is achieved through mechanical transparency and motor-current-based torque sensing rather than external force sensors."

Technical:

"Because Neo's joints are quasi-direct and tendon-driven, joint torque propagates directly to the motor, enabling high-bandwidth proprioceptive force estimation and active impedance control."

Biomechanics framing:

"Neo replaces physical tendons and Golgi organs with a transparent actuator and software reflexes, achieving muscle-like force sensing."

1. First: Reset the Definition (This Is Where Confusion Started)

What proprioception actually is (core definition)

Proprioception is the ability of a system to estimate its own mechanical state using internal sensing alone.

That state includes:

- joint position
- joint velocity
- joint torque / force

! Proprioception is **not** about *what* variables you know
! It is about **where that information comes from**

In robots:

- The **motor is the muscle**
- If the motor does not feel load → the robot has no "muscle sense"

So when we say:

"External forces must propagate back to the actuator"

we mean:

Joint interaction forces must be observable at the actuator level

Otherwise:

- the actuator is "blind"
- control becomes feedforward or delayed
- compliance becomes unsafe or sluggish

----->>

Powered Knee:

Mechanical Transparency of Your Knee

Ball screw properties:

- Low friction (especially if well-lubricated)
- Mechanically backdriveable
- Force at joint → axial force → motor torque

So yes:

Joint interaction forces *can* propagate back to the motor

That is important. "The system was mechanically transparent and could support proprioceptive control."

That is a **very strong and honest statement**.

Why internal force sensing matters

1. Scalability

- You cannot put force sensors in:
 - every joint
 - every finger
 - every tendon

Humanoids need **dozens of force channels**.

2. Bandwidth

- Motor current sensing is:
 - kHz-level
 - low latency
- Force sensors:
 - noisier
 - lower bandwidth
 - often fragile

7. Joint Torque vs Motor Torque — This Is the Core Equation

You asked:

"Joint torque or motor torque — which one matters?"

The answer:

Joint torque matters.

Motor torque is how we estimate it.

Proprioception works when:

$$\tau_{\text{joint}} = r \cdot \tau_{\text{motor}} \Rightarrow \tau_{\text{motor}} = \tau_{\text{joint}} / r$$

Where:

- r is known transmission ratio
- losses are small and predictable

This requires:

- low friction
- low backlash
- transparency

Three layers

- Mechanical transparency**
→ enables force to propagate
- Internal force estimation**
→ motor current → torque
- Active impedance control**
→ software-defined compliance

Only when **all three** exist:

→ the system is proprioceptive

Soft Finger Tip

1. How Active Inflatable Fingertips Work

The system consists of a rigid "bone" (distal phalanx), a flexible elastomeric "skin," and an internal air chamber.

- **Mechanism:** A miniature pump (pneumatic) or a micro-piston (hydraulic) forces fluid into the fingertip. This changes the internal pressure (P) and the curvature of the skin.
- **The Material Analog:** The skin is usually a **Viscoelastic Elastomer** (mimicking the "Elastin" and "Fatty Tissue" in human fingertips discussed in your notes).

2. Why Inflate First, then Squeeze? (The Physics of "Grip")

If you have a force sensor, you might think you can just squeeze. However, inflation is about **Contact Geometry**, not just force.

- **Hertzian Contact Pressure:** When a rigid finger touches a rigid object, the contact area is a "point." This results in extremely high local pressure, which can damage the item or cause it to "squirt" out of the hand.
- **Conformity (The Goal):** By inflating *before* the squeeze, you increase the **Contact Surface Area (A)**.
- **The Friction Benefit:** Friction force is $F_f = \mu F_n$. While μ (the coefficient of friction) is a material property, in soft materials, the "effective friction" increases as the skin "interlocks" with the micro-texture of the object.
- **The Sequence:**
 - a. **Pre-Inflation:** Softens the finger to "wrap" around the object's geometry.
 - b. **Contact Detection:** The robot senses the moment of touch.
 - c. **The Squeeze:** The cable-driven tendons apply the high-tension force to "lock" the object in place.

3. Sensing Strategy: External vs. Proprioception

You asked if we should use external sensors or 1X-style proprioception. The answer for a humanoid is usually **both**, but used differently:

A. Internal Pneumatic Sensing (The "Active" Sensor)

When the inflatable fingertip touches an object, the air inside is compressed. By placing a **Pressure Sensor** in the air line (inside the palm/arm), you can detect the contact without putting fragile electronics on the fingertip.

- **Advantage:** Extremely robust. No wires at the fingertip to break.
- **Data:** It provides "Area-Averaged" force.

B. Cable-Driven Proprioception (1X Style)

Once the finger is squeezed, the **Motor Current/Tendon Tension** tells you the "Global Grip Force."

- **Synergy:** Use the **Pneumatic Pressure** to detect "Initial Touch" and "Slip," and use **Tendon Proprioception** to control the "Crushing Force."

4. Implementation on a Humanoid (Neo)

In prosthetics, these are often "dumb" (always inflated). In a humanoid, they must be "smart."

- **The Source of Inflation:**
 - * **Micro-Bellows or Pumps:** Small DC diaphragm pumps located in the forearm.
 - **Shared Manifold:** A single pump in the torso with small solenoid valves in the hand to "address" each finger individually.
- **The "Skin" Design:**
 - The cover should be **Anisotropic** (like your bone/tendon notes). It should be stretchy at the tip but reinforced with fibers on the sides so it doesn't "balloon" out sideways, only forward toward the object.

5. Why this is better than "Just Carbon Fiber"

If you tell Christophe you want a "Carbon Fiber shell," he will ask how you handle an egg or a wine glass. Carbon fiber has zero **Passive Compliance**.

- **Energy Absorption:** Like the "Spongy Bone" in your notes (B4), the inflatable tip acts as a shock absorber.
- **In-Hand Manipulation:** If Neo wants to "roll" a pen between its fingers, the inflatable tips can shift their "center of pressure" by changing inflation levels, allowing for dexterity that rigid fingers cannot achieve.

6. The Sequence of Operation (The "Script")

If Christophe asks how you would program a grip:

1. **Approach:** Proprioception moves the hand near the object.
2. **Pre-Shape:** Inflate fingertips to **Low Pressure** (P_{low}) to maximize potential contact area.
3. **Touch:** Monitor **Internal Air Pressure** for a spike, signifying contact.
4. **Conform:** Increase inflation to **High Pressure** (P_{high}) to "wrap" the object.
5. **Grasp:** Engage the **Tendon Actuators** (Cable-Driven) to apply the required N Newtons of force.
6. **Monitor:** If the Air Pressure fluctuates rapidly, it indicates **Micro-Slip**; the controller then increases tendon tension.

Mechanical Design:

1. The "15-Year" Question: Fatigue and Failure Analysis

When an interviewer asks, "What will break after 15 years?" they are testing your knowledge of **Fatigue Life** and **Wear**.

The Concept: Fatigue (Cyclic Loading)

Most robot parts don't break because a force was too big (Static Failure); they break because a medium force was applied 10 million times (Fatigue Failure).

- **The Formula:** You should mention the **S-N Curve** (Stress vs. Number of Cycles).
- **The Equation:** Use the **Goodman Criterion** or **Gerber Criterion** to calculate the factor of safety for a part experiencing "Mean Stress" (the weight of the robot) and "Alternating Stress" (the vibration/impact of walking).

$$S_e / S_{mean} + S_{alt} / S_{ut} = 1/n$$

(Where S_e is the endurance limit—the stress level below which a material can theoretically last forever).

What is the "Critical Part"?

In your ankle prosthesis or a humanoid joint, the most vulnerable parts are:

1. **Bearings (Surface Pitting):** High-impact walking causes "rolling contact fatigue." The metal surfaces start to flake off at a microscopic level.
2. **Seals (The Hydraulic "Lung"):** In your hydraulic unit, the **Dynamic Seals** (rubbing against the piston) are the #1 failure point. Friction wears the polymer, leading to leaks.
3. **Springs (Stress Relaxation):** Over 15 years, a spring under constant load will "lose its memory" and won't return to its original length.
4. **Cams/Ropes (Stress Concentration):** In the 1X tendon system or your clutch, the point where the rope bends over a small radius is the "Stress Concentration" point.

2. Hand Calculations vs. Simulation: Why Hand Calcs Matter

The interviewer's comment "Simulation always gives a result" is a trap. It's called **GIGO (Garbage In, Garbage Out)**.

The Answer: *"I use hand calculations as a **Sanity Check**. A simulation can tell me the stress is 200 MPa, but if my hand calculation (Force / Area) tells me it should be 20 MPa, I know I set up my simulation constraints or mesh incorrectly. Hand calculations allow me to 'size' the components (decide if a shaft should be 10mm or 20mm) before I ever open CAD. It saves time and prevents 'over-engineering'."*

3. The "Simple Button" Design Process

This question is about **Systems Engineering**. They want to see if you have a repeatable "Recipe" for design.

The Block Diagram for a Button (or an Ankle):

1. **Requirements (The 'What'):** How many clicks must it survive? (Cycle life). How much force to press? (Tactile feedback). Environment? (Dust/Waterproof - IP rating).
2. **Functional Decomposition:** It needs an **Actuator** (the cap), a **Return Mechanism** (the spring), and an **Interface** (the electrical contact).
3. **Material Selection:** Do I use Stainless Steel for the spring (fatigue resistance) or Plastic for the housing (low cost/weight)?
4. **Failure Mode & Effects Analysis (FMEA):** What if the spring snaps? Does the button stay "on" (Dangerous) or "off" (Safe)?
5. **Prototyping & DFM (Design for Manufacturing):** Can I make 10,000 of these cheaply using injection molding?

Using **7075 Aluminum** (specifically the **T6 temper**) is a hallmark of high-performance robotics and aerospace engineering. In a biomechanics interview, your choice of this material proves that you understand the trade-offs between **mass, inertia, and cyclic fatigue**.

Here is the technical breakdown of why 7075 was the correct choice for your hybrid ankle prosthesis and how to explain it to Christophe.

1. Superior Strength-to-Weight Ratio (Specific Strength)

The most critical requirement for a prosthesis is that it must be light enough for an amputee to swing without fatigue, yet strong enough to support 3–5 times their body weight during an impact (heel strike).

- **The Advantage:** 7075-T6 has a yield strength of approximately **500 MPa**, which is comparable to many structural steels. However, aluminum is **one-third the density** of steel.
- **Why you used it:** By using 7075, you reduced the **"Swing Mass"** of the prosthesis. Lower mass at the ankle reduces the metabolic cost for the user and minimizes the "pendulum effect," making the gait feel more natural and aligned with 1X's goal of "human-like" motion.

2. High Fatigue Strength

As we discussed with the "15-year" question, a prosthesis is a **cyclic load device**. It will experience millions of steps.

- **The Advantage:** While most aluminum alloys have poor fatigue limits (meaning they will eventually fail even under small loads), 7075 has one of the highest fatigue strengths of any aluminum series.
- **Why you used it:** You needed a material that could handle the **Stochastic Loading** (unpredictable forces) of real-world walking without developing microscopic cracks at the stress concentration points of your clutch or hydraulic housing.

3. Machinability for Complex Bio-Inspired Geometries

Your design includes a "passive clutch" with a "cam-railway geometry." These are complex parts that require high precision.

- **The Advantage:** Unlike titanium (which is difficult and expensive to machine) or cast metals, 7075-T6 machines beautifully. It holds tight tolerances and doesn't "gum up" the cutting tools.
- **Why you used it:** It allowed you to iterate on the **Complex Mechanical Logic** of your clutch mechanism quickly while ensuring that the internal surfaces of your hydraulic cylinder were smooth enough for high-performance seals.

Looking at your **B4 - Bone Biomechanics** file, bone is described as having a high modulus of elasticity but being susceptible to **Fatigue Fractures** when the "rate of damage exceeds the rate of remodeling" (page 41).

- **Deep Insight for the Interview:** *"Biological bone is a self-healing composite. Since my prosthesis is 'In-Vitro' (non-living), I chose 7075 Aluminum because its **Endurance Limit** provides a mechanical buffer against the fatigue failure that biological systems solve through constant remodeling. I am essentially over-engineering the material properties to compensate for the lack of biological repair."*

ART I — Why they asked you those questions (the meta point)

When a mechanical interviewer asks:

"If I come back after 15 years, what will fail first?"

They are testing whether you:

- Understand **load paths**
- Think in terms of **fatigue, wear, tolerances, and interfaces**
- Know the difference between **strength vs durability**
- Can reason *before* simulation

Simulation answers alone often signal:

"I ran tools, but I didn't reason physically first."

That's why they pushed you on **hand calculations**.

PART II — The core mechanical design mindset (this is the key)

Mechanical design is **not** about geometry first.

It is about **forces → load paths → failure modes → mitigation**.

Every design should follow this mental sequence:

1. **What loads exist?**
2. **Where do those loads flow?**
3. **Which components see cyclic vs static load?**
4. **Which interfaces amplify stress?**
5. **What fails by fatigue, wear, creep, or leakage?**

Simulation is used after this reasoning, not instead of it.

PART III — Simple example: Designing a button (what Flexiv was asking)

Let's do this properly.

Step 1 — Define the function

A button must:

- Be pressed repeatedly
- Provide tactile feedback
- Return reliably
- Survive millions of cycles

Already, you should think: **fatigue**, not strength.

Step 2 — Identify loads

Typical button loads:

- Force: ~1–5 N per press
- Cyclic loading: 10⁵–10⁷ cycles
- Local contact stress at interfaces

This is **low force, high cycle** → classic fatigue & wear problem.

Step 3 — Identify load path

Finger → button cap → spring → housing → frame

Critical insight:

The weakest point is *not* the spring force—it's the **contact interfaces**.

Step 4 — Identify failure modes

Likely failures:

- Spring fatigue (loss of preload)
- Plastic creep in housing
- Wear at contact surfaces
- Loss of tactile click

Step 5 — Hand calculations (why they matter)

For the spring:

- Use **S–N curves**
- Goodman or Soderberg criteria
- Check endurance limit

For contacts:

- Hertzian contact stress
- Wear rate (Archard's law)

You don't need exact numbers in interview—what matters is that you **mention the reasoning**.

How to answer in interview (button question)

"I'd start by identifying cyclic loads and load paths. For a button, fatigue and wear dominate rather than static strength. I would estimate spring stress using S–N curves, evaluate contact stresses at interfaces, and then use simulation to validate assumptions and identify stress concentrations."

That answer alone passes.

PART IV — Now scale this to your ankle prosthesis

This is where your project actually shines.

Step 1 — Load characterization (this is where you should start)

An ankle prosthesis experiences:

- **High cyclic loads** (1–2 million steps/year)
 - Ground reaction forces ~1–1.3× body weight
 - Shock loads at heel strike
 - Direction-dependent loading
- So immediately:
- Fatigue, wear, sealing, and friction** dominate—not yield failure.
- Step 2 — Load paths (CRITICAL)**
- In your ankle:
- Ground → carbon foot → ankle joint → hydraulic pistons → frame → pylon
- The **critical insight**:
- Loads do not distribute evenly — they concentrate at interfaces.
- Step 3 — Identify critical components (this is what Agility wanted)**
- If someone asked:
- "What fails after 15 years?"
- A strong answer would be:
- Most vulnerable components (ranked)**
1. **Seals in hydraulic pistons**
 - Cyclic pressure
 - Wear + leakage
 1. **Cam / clutch interface surfaces**
 - Sliding contact
 - Fretting and surface fatigue
 1. **Spring anchoring points**
 - Stress concentration
 - Fatigue crack initiation
 1. **Carbon foot–frame interface**
 - Delamination risk
 1. **Bearings or bushings**
 - Lubrication loss

Notice:

You didn't say "the spring breaks" — because springs usually survive if designed properly.

Step 4 — Why hand calculations matter here

Before Abaqus, you should be able to say:

- Estimated ankle torque:

$$\tau = FGRF \times r_{GRF} = F_{GRF} \times r$$

- Estimate piston force:

$$F = \tau / r_{piston} = \frac{\tau}{r_{piston}} = F_{piston}$$

- Estimate contact pressure:

$$\sigma = F / A$$

Even **order-of-magnitude** calculations show:

"I understand the physics."

Simulation refines—not replaces—this.

Step 5 — Fatigue vs strength (this is where you missed earlier)

Key distinction:

- **Strength** → "Does it break once?"
- **Fatigue** → "Does it survive 10 million times?"

Most prostheses fail by:

- Micro-cracks
- Seal wear
- Loss of damping
- Increased backlash

Not by catastrophic fracture.

How to answer the "15-year" question (use this verbatim)

"I would expect wear- and fatigue-driven failures rather than static fracture. In this design, the most vulnerable components are hydraulic seals, sliding clutch interfaces, and spring anchor points due to cyclic loading. I addressed this by minimizing stress concentrations, selecting fatigue-rated materials, and validating load paths through both hand calculations and dynamic simulations."

That is a **senior-level answer**.

Positive Force Feedback

2. What is Positive Force Feedback (PFF), really?

This is the core conceptual confusion—let’s fix it properly.

The three major muscle sensory pathways

In human physiology, muscles are *not* controlled only by position or velocity.

They have **three distinct feedback signals**:

(A) Muscle spindle — length & velocity feedback

Sensors: Muscle spindles

Measures:

- Muscle length
- Muscle velocity

Function:

- Posture control
- Stretch reflex
- Stabilization against perturbations

Control effect:

- Resists changes in length
- Acts like stiffness and damping

☞ This is analogous to **impedance control**

(B) Golgi tendon organ (GTO) — force feedback

Sensors: Golgi tendon organs

Measures:

- Muscle force / tension

This is where **Positive Force Feedback (PFF)** comes from.

(C) Central command (activation)

Source: CNS / EMG

Measures:

- Intended muscle activation

3. What does *positive* force feedback mean?

This is subtle but extremely important.

Classical intuition (wrong but common)

People think feedback should always be **negative** for stability.

But **locomotion is not static control**.

In walking, PFF does this:

When muscle force increases, activation is increased further—temporarily

That sounds unstable, but it’s phase-dependent.

Why PFF exists biologically

During stance and push-off:

- The body *wants* force to build rapidly
- The ankle plantarflexors (soleus, gastrocnemius) **amplify force**
- PFF helps:
 - Rapid load acceptance
 - Efficient energy transfer
 - Push-off power generation

This is **functional amplification**, not instability.

4. Why not just length or velocity feedback?

Excellent question.

Length / velocity feedback:

- Good for **stability**
- Good for **disturbance rejection**
- Bad for **power amplification**

Force feedback:

- Directly reflects **interaction with the environment**
- Scales with load
- Enables **emergent behavior** (stronger push when more load exists)

In other words:

Length/velocity feedback stabilizes posture

Force feedback enables propulsion

Your neuromuscular ankle controller is exploiting exactly this principle.

1. What positive force feedback means in biology (ground truth)

In human physiology:

- **Golgi tendon organs (GTOs)** measure **muscle force**
- Their output is fed back to the **motor neuron pool**
- During locomotion (stance & push-off), this feedback is **positive**

That means:

An increase in muscle force causes an *increase* in muscle activation, which further increases muscle force.

This sounds unstable—but it’s not, because:

- It is **phase-gated**
- It is **time-limited**
- It is **embedded in a compliant musculoskeletal system**

This loop exists **only when load-bearing is desired**.

2. What is the feedback signal in PFF?

This is the key answer to your question:

The feedback signal is muscle force (or tendon force).

Not length.

Not velocity.

Force.

In biology:

- Force is sensed **directly** by GTOs in the tendon.

3. Where is the feedback loop closed mathematically?

Let’s write a simplified neuromuscular model.

Standard Hill-type structure

Muscle force:

$$F_m = a \cdot F_{max} \cdot f_l(l) \cdot f_v(\dot{l})$$
$$F_m = a \cdot F_{max} \cdot f_l(l) \cdot f_v(\dot{l})$$

Where:

- a = activation (from EMG)
- f_l = force–length relation
- f_v = force–velocity relation

So far: **no feedback** — this is feedforward.

Now add positive force feedback

Activation is **not just EMG** anymore.

Instead:

$$a(t) = a_{EMG}(t) + k_f \cdot F_m(t - \tau)$$
$$a(t) = a_{EMG}(t) + k_f \cdot F_m(t - \tau)$$

This is the key equation.

☞ **THIS is positive force feedback**

- The force generated at time $t - \tau$ is fed back
- It increases activation at time t
- This closes the loop

Biologically:

- k_f = reflex gain
- τ = neural delay (~30–50 ms)

4. Where does force come from in a robot?

This is where your confusion is totally valid.

In a robot, force can come from:

1. **Direct force/torque sensor**
 - Load cell
 - Strain gauge
1. **Motor current**
 - $\tau \propto i$
1. **Estimated force**
 - From dynamics
 - From motor + kinematics
 - From impedance model

All are valid, but **they are not equivalent** conceptually.

5. Now let’s be very honest about *your* implementation

You said:

“We are using EMG activation, estimating force–length and force–velocity relations, identifying max force, applying activation, and then driving the actuator.”

That means:

What you definitely have:

- EMG-based **feedforward activation**
- Muscle-inspired **force generation**
- Neuromuscular **actuation model**

The critical question:

Is muscle force *fed back* into activation?

Let’s check.

Case A — No force feedback loop (most common)

If your controller is:

$$a = a_{EMG}$$
$$F = f(a, l, \dot{l})$$

Then:

✗ **This is NOT positive force feedback**

☑ **This is neuromuscular feedforward control**

Still bio-inspired — but **not reflexive PFF**.

Case B — Force modifies activation (true PFF)

If your controller does:

$$a(t) = a_{EMG}(t) + k_f \cdot F(t - \tau)$$
$$a(t) = a_{EMG}(t) + k_f \cdot F(t - \tau)$$

Where F is:

- measured torque
- motor current
- estimated tendon force

Then:

☑ **This IS positive force feedback**

The loop is closed on **force**.

6. So where does the feedback come from?

Let me answer your question **directly**:

“Where is the feedback coming from?”

Correct answers (depending on implementation):

- From **measured joint torque**
- From **motor current**
- From **estimated muscle/tendon force**

Incorrect answer:

- From EMG alone (EMG is command, not feedback)

7. Why force feedback matters (deep insight)

Length & velocity feedback answer:

“Where am I and how fast am I moving?”

Force feedback answers:

“How hard am I interacting with the world?”

That is why:

- PFF scales torque with **load**
- It enables **emergent push-off**
- It avoids rigid trajectory enforcement

8. How you should describe *your* project accurately

Here is the **safe and correct phrasing**, depending on what you actually implemented.

If you did NOT close force feedback:

“I implemented a neuromuscular ankle controller based on EMG-driven activation and muscle force generation, inspired by reflex-based control, but without closing an explicit force feedback loop.”

If you DID close force feedback (even weakly):

“I incorporated a reflex-inspired positive force feedback term, where estimated joint torque modulated activation to enable load-dependent ankle assistance.”

Never claim PFF unless the loop is explicit.

9. One-sentence mental model (remember this)

Positive force feedback exists only if force changes activation.

If force does not affect activation, it’s not feedback.

4. Translating Human Actuation Biomechanics into Robot Design

How it works in humans

Human actuation is not stiff torque sources.

Key properties:

- Muscles generate force through **compliant tendons**
- Force depends on **length, velocity, and activation**
- Reflexes provide fast, local correction
- Voluntary control operates at slower timescales

Humans combine:

- Passive mechanics
- Reflexive responses
- Volitional intent

What matters biomechanically

- Compliance for shock absorption
- Energy storage and return
- Bandwidth separation (fast reflex, slow intent)
- Stability through interaction, not dominance

How I would translate this to robot design

I would **not directly copy muscles**, but capture their **functional role**:

1. **Compliant actuation**
 - SEA, PEA, tendon-driven systems
 - Mechanical transparency
1. **Impedance-based control**
 - Adjustable stiffness and damping
 - Yield to external forces safely
1. **Reflex-inspired control**
 - Fast local responses to deviations
 - Supervisory safety layers (your FTM)
1. **Shared control**
 - Autonomous motion + human intent
 - Similar to CPG + voluntary override

Connection to your work

- Passive ankle prosthesis
- Neuromuscular models
- Shared control and impedance modulation
- Reflex-inspired fault tolerance

“I design actuation systems that cooperate with the human rather than overpower them.”

3. Translating Human Sensing Biomechanics into Robot Design

How it works in humans

Humans sense motion and interaction through:

- **Muscle spindles** → length & velocity
- **Golgi tendon organs** → force
- **Cutaneous receptors** → pressure & shear
- **Vestibular system** → acceleration & orientation

Important insight:

Humans do **not measure everything precisely**.

They sense **what matters for stability and safety**.

Sensing is:

- Distributed
- Redundant
- Context-dependent

What matters biomechanically

- Relative motion, not absolute precision
- Rate of change (velocity, acceleration)
- Threshold-based responses
- Integration across modalities

How I would translate this to robot design

I would focus on **proprioceptive and interaction-relevant sensing**:

1. **Joint-level proprioception**
 - Motor current → torque
 - Encoders → position & velocity
1. **Contact sensing**
 - Distributed pressure sensors
 - Early detection of unexpected contact
1. **State estimation**
 - Compare expected vs measured motion
 - Detect abnormal interaction (your FTM work)
1. **Design sensing to serve control**
 - Not maximum resolution
 - Sufficient resolution at correct bandwidth

Connection to your work

- Internal model vs measured motion
- Velocity-based error detection
- Proprioceptive control without heavy external sensing

“I design sensing to support stability and safety, not just measurement.”

2. Translating Human Skin Biomechanics into Robot Design

How it works in humans

Human skin is:

- **Layered** (epidermis, dermis, fat)
- **Viscoelastic**
- **Highly deformable**
- **A mechanical + sensory interface**

Key functions:

- Distributes contact forces
- Absorbs impact energy
- Prevents sharp pressure peaks
- Houses dense tactile sensing
- Enables early detection of contact

Skin does **not stop force** — it **spreads it in time and space**

What matters biomechanically

- Pressure distribution
- Rate-dependent response (fast vs slow contact)
- Contact area growth with load
- Coupling between deformation and sensing

How I would translate this to robot design

For robot skin, I would think in **layers**, just like biology:

1. **Outer compliant layer**
 - Soft elastomer or lattice
 - First line of energy absorption
 - Prevents high contact stress
1. **Intermediate damping layer**
 - Viscoelastic foam or structured polymer
 - Absorbs impact energy
 - Controls force rise rate
1. **Inner sensing layer**
 - Pressure or force sensors
 - Detect distributed contact, not just point force
1. **Design targets derived from humans**
 - Maximum safe pressure
 - Acceptable force rise time
 - Deformation vs force curves

Connection to your work

- Insole pressure sensing as cutaneous analog
- Impact absorption in ankle prosthesis
- Hydraulic damping as soft tissue analog

“I would design robot skin as a mechanical signal conditioner that protects both the human and the robot before control reacts.”

1. Translating Human Bone Biomechanics into Robot Design

How it works in humans

Human bones are not rigid sticks. They are:

- **Load-bearing structures** with **anisotropic stiffness**
- Optimized for **bending, torsion, and compression**
- Designed to **redirect forces**, not just resist them

Key properties:

- Cortical bone: stiff, load-carrying
- Trabecular bone: compliant, energy-dissipating
- Geometry (cross-section, curvature) matters more than material strength alone
- Bones work with joints to **shape force paths** during walking, impacts, and falls

Bones are also part of a **hierarchical system**:

- Bone geometry → joint kinematics → muscle leverage → whole-body dynamics

What matters biomechanically

- **Directional stiffness** (strong where needed, compliant where safe)
- **Moment arms and leverage**
- **Load paths under dynamic motion**
- **Stress distribution over time (fatigue)**

How I would translate this to robot design

In robots, I would **not try to replicate bone anatomy**, but rather:

1. **Use geometry to control stiffness**
 - Vary cross-sections to tune bending vs torsional stiffness
 - Use curved or tapered links to redirect loads safely
1. **Design structural compliance intentionally**
 - Embed compliance in links or joints where humans tolerate motion
 - Keep stiffness high where precision is required
1. **Design for fatigue, not just peak load**
 - Bones survive millions of cycles → robots must too
 - Use fatigue-informed design (not only static FEA)
1. **Exploit structure before control**
 - Let mechanical geometry do part of the stabilization
 - Reduce reliance on high-bandwidth control loops

Connection to your work

You can directly reference:

- Slider-crank prosthesis geometry
- Passive ankle prosthesis structural layout
- Load redirection through linkage design

“I think of robot ‘bones’ as load-guiding structures that shape forces before control even starts.”